

scpviz: A Python bioinformatics toolkit for Single-cell Proteomics and multi-omics analysis

Marion Pang¹, Baiyi Quan², Ting-Yu Wang², and Tsui-Fen Chou^{1,2}

¹ Division of Biology and Biological Engineering, California Institute of Technology, 1200 E. California Blvd, Pasadena, CA 91125 ² Proteome Exploration Laboratory, Beckman Institute, California Institute of Technology, 1200 E. California Blvd, Pasadena, CA 91125 ¶ Corresponding author

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Summary

Proteomics seeks to characterize protein dynamics by measuring both protein abundance and post-translational modifications (PTMs), such as phosphorylation, acetylation, and ubiquitination, which regulate protein activity, localization, and interactions. In bottom-up proteomics workflows, proteins are enzymatically digested into peptides that are measured as spectra, from which these peptide-spectrum matches (PSMs) are aggregated to infer protein-level identifications and quantitative abundance estimates. Analyzing the two levels of data at both the peptide level (short fragments observed directly) and the protein level (assembled from peptide evidence) in tandem is crucial for translating raw measurements into biologically interpretable results.

Single-cell proteomics extends these approaches to resolve protein expression at the level of individual cells or microdissected tissue regions. Such data are typically sparse, with many missing values, and are generated within complex experimental designs involving multiple classes of samples (e.g., cell type, treatment, condition). These properties distinguish single-cell proteomics from bulk experiments and create unique challenges in data processing, normalization, and interpretation. The single-cell transcriptomics community has established a mature ecosystem for managing similar challenges, exemplified by the scanpy package (Wolf et al., 2018) and the broader scverse ecosystem (Virshup et al., 2023). Building on these foundations, scpviz extends the AnnData data structure to the domain of proteomics, supporting a complete analysis pipeline from raw peptide-level data to protein-level summaries and downstream interpretation through differential expression, enrichment analysis, and network analysis. The core of scpviz is the pAnnData class, an AnnData-affiliated data structure specialized for proteomics. Together, these components make scpviz a comprehensive and extensible framework for single-cell proteomics. By combining flexible data structures, reproducible workflows, and seamless integration with the AnnData, scanpy and extended scverse ecosystem, the package enables researchers to efficiently connect peptide-level evidence to protein-level interpretation, thereby accelerating methodological development and biological discovery in proteomics.

Statement of need

Although general-purpose data analysis frameworks such as scanpy (Wolf et al., 2018) and the broader scverse ecosystem have become indispensable for single-cell transcriptomics, comparable tools for proteomics remain limited. Unlike transcriptomic data (counts, reads), proteomics measurements are inherently hierarchical. Peptide-level measurements provide the primary basis from which protein-level quantities are inferred from, often with shared

or ambiguous peptide assignments and strong reliance on peptide confidence. Compounded to these properties is the problem of missing data in low-input and single-cell experiments, which make proteomics data difficult to represent and analyze using flat feature-by-sample abstractions.

Existing proteomics software typically focuses on upstream tasks such as peptide identification and protein quantification, producing tabular outputs that are not designed for iterative downstream analysis. As a result, users must rely on ad hoc data structures to perform filtering, normalization, visualization, and interpretation, limiting reproducibility and making it difficult to integrate proteomics data into modern single-cell analysis workflows. These limitations are amplified in single-cell and spatial contexts, where complex experimental designs, sparse measurements, and the need to compare across multiple biological conditions require flexible data management and metadata-aware analysis.

scpviz addresses this gap by providing a unified framework system for organizing and analyzing proteomics data from raw peptide-level evidence through protein-level summaries and biological interpretation. It is designed for computational biologists and proteomics researchers working with low-input or single-cell datasets generated by common analysis pipelines such as Proteome Discoverer or DIA-NN (Demichev et al., 2020). By enabling structured downstream analysis and integration with established single-cell ecosystems, scpviz supports reproducible and scalable proteomics workflows and facilitates cross-modality analyses that connect protein-level measurements to broader systems-level biology.

Software design

The design of scpviz centers on the pAnnData class, an AnnData-affiliated data structure specialized for proteomics. Rather than representing proteomics data as a single flat matrix, pAnnData accounts for the hierarchical relationship between peptide-level and protein-level measurements by pairing matched peptide (.pep) and protein (.prot) AnnData objects with supporting attributes such as .summary, .metadata, .stats, and a protein-peptide relationship structure (.rs matrix). This design allows users to preserve explicit peptide-protein relationships during downstream analyses, enabling operations like peptide-level protein abundance normalization or peptide-based fold-change aggregation for differential expression, while maintaining compatibility with established Python libraries for data science and visualization.

Proteomics-specific operations in scpviz are designed to operate uniformly across peptide and protein-level data. By organizing functionality into mixin-based classes that act on underlying AnnData objects, common operations such as filtering, normalization, imputation, and summarization can be applied consistently to both peptides and proteins. This object-oriented design reduces code duplication while ensuring that shared analytical logic respects proteomics-specific constraints at each level of representation. Visualization and analysis utilities (e.g. PCA, UMAP, clustermaps, abundance plots) build directly on these structured representations (McInnes et al., 2018). For downstream interpretation, scpviz integrates external resources such as UniProt for annotation and STRING database for functional enrichment and network analysis (Snel et al., 2000; Szklarczyk et al., 2023), and incorporates proteomics-specific quantification strategies such as directLFQ (Ammar et al., 2023). By retaining AnnData compatibility, pAnnData objects can be used directly with tools such as scanpy (Wolf et al., 2018) and harmony (Korsunsky et al., 2019), enabling direct incorporation into established single-cell workflows.

The decision to develop scpviz as a new package, rather than contributing to existing single-cell or proteomics frameworks, reflects a structural gap between these ecosystems. Current single-cell frameworks are designed around single-feature matrices and lack native support for hierarchical molecular evidence, while existing proteomics tools focus on identification and quantification rather than downstream, metadata-aware analysis. Embedding proteomics-

specific logic directly into AnnData or extending upstream proteomics software would either constrain their generality or fail to support iterative single-cell analyses. scpviz bridges this gap by extending, rather than replacing, AnnData, providing a dedicated data model that preserves proteomics-specific structure while enabling direct reuse of mature single-cell analysis infrastructure.

The design philosophy of scpviz emphasizes both usability and extensibility. General users can rely on its streamlined API to import, process, and visualize single-cell proteomics data without deep programming expertise, while advanced users can extend the framework to accommodate custom analysis pipelines.

Research impact statement

scpviz has already been used in the analysis of multiple published studies and preprints across single-cell and bulk proteomics applications (Dutta, Pang, Coughlin, et al., 2025; Dutta, Pang, Donahue, et al., 2025; Pang et al., 2025; Uslan et al., 2025). In these works, the framework enabled structured downstream analysis of peptide and protein-level data, including differential expression, functional enrichment, and integration with transcriptomic measurements. In addition, scpviz has been incorporated into graduate-level training to demonstrate how proteomics workflows can be analyzed using architectures common in single-cell transcriptomics analysis, lowering the barrier for new users entering the field.

The primary impact of scpviz lies in providing reusable infrastructure rather than task-specific analyses. By formalizing peptide–protein relationships within an AnnData-compatible data model, the package enables proteomics data to be analyzed using established single-cell tools for visualization, integration, and exploratory analysis. This structured representation supports reproducible downstream workflows and facilitates multi-omics studies in which proteomics and transcriptomics data can be jointly interpreted, enabling analyses of protein abundance dynamics, signaling pathways, and cellular heterogeneity in single-cell and spatial proteomics experiments.

scpviz is released as open-source software with comprehensive documentation, automated tests, and reproducible examples, supporting transparent and extensible research workflows. Its emphasis on interoperability with widely used single-cell analysis libraries positions it as infrastructure for continued method development as single-cell and spatial proteomics technologies mature and scale.

AI usage disclosure

Generative AI tools were used during the development of this work to assist with code refactoring, documentation drafting, and manuscript text editing. All software design decisions, implementation, validation, and scientific interpretation were performed and reviewed by the authors. No generative AI tools were used to generate or analyze research data, and all results reported are reproducible from the publicly available source code and documentation.

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